GFN
Microsoft Azure 5

Tag 14
09.07.2026
Donnerstag

Von KI erstellt, von Menschen kontrolliert.

Guten Morgen!

GitHub Repo https://github.com/www42/a5

AZ-104

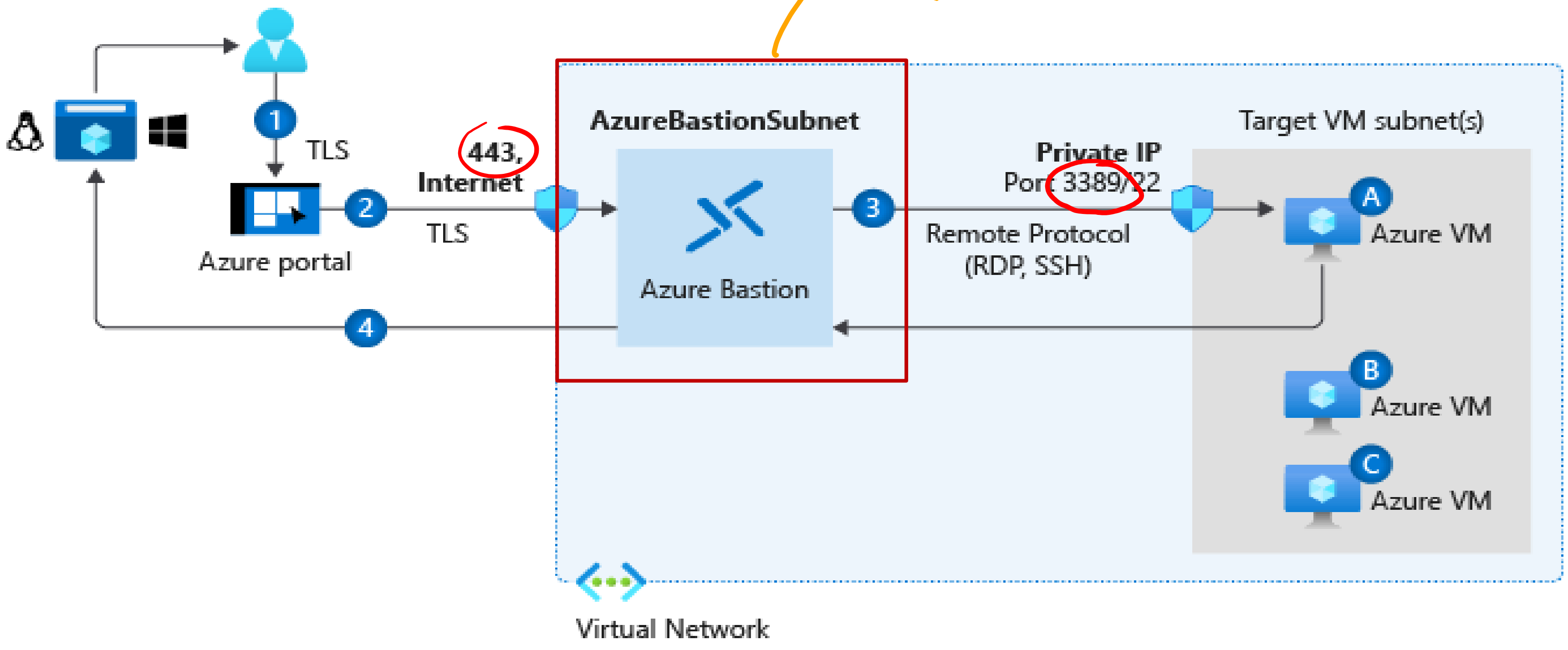
Learning Paths:

- ➡️ Manage identities and governance in Azure
- ➡️ Prerequisites for Azure administrators – CloudShell and templates
- ➡️ Configure and manage virtual networks for Azure administrators
- ➡️ Implement and manage storage in Azure
- ➡️ Deploy and manage Azure compute resources
- ➡️ Monitor and back up Azure resources

What happened yesterday

Connect to Virtual Machines

Student SKU (Free)
(Developer?)



Bastion Subnet for RDP/SSH through the Portal over SSL

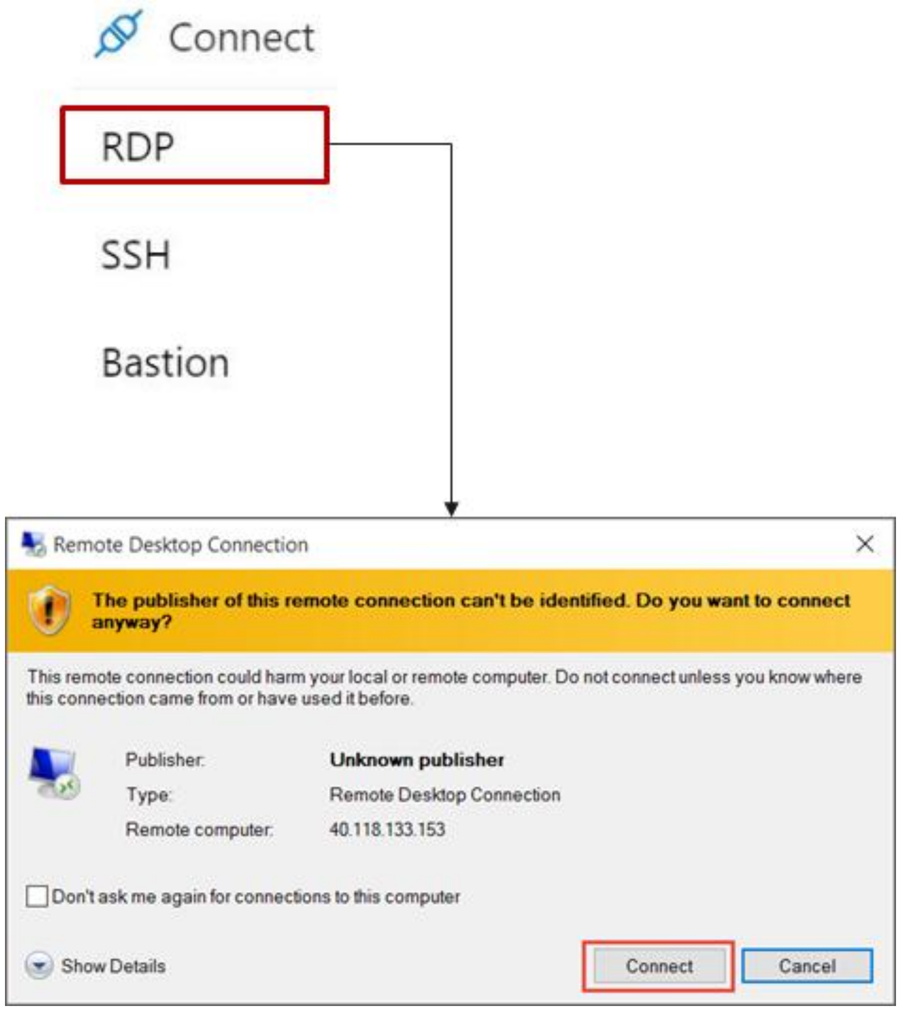
Remote Desktop Protocol for Windows-based Virtual Machines

Secure Shell Protocol for Linux based Virtual Machines

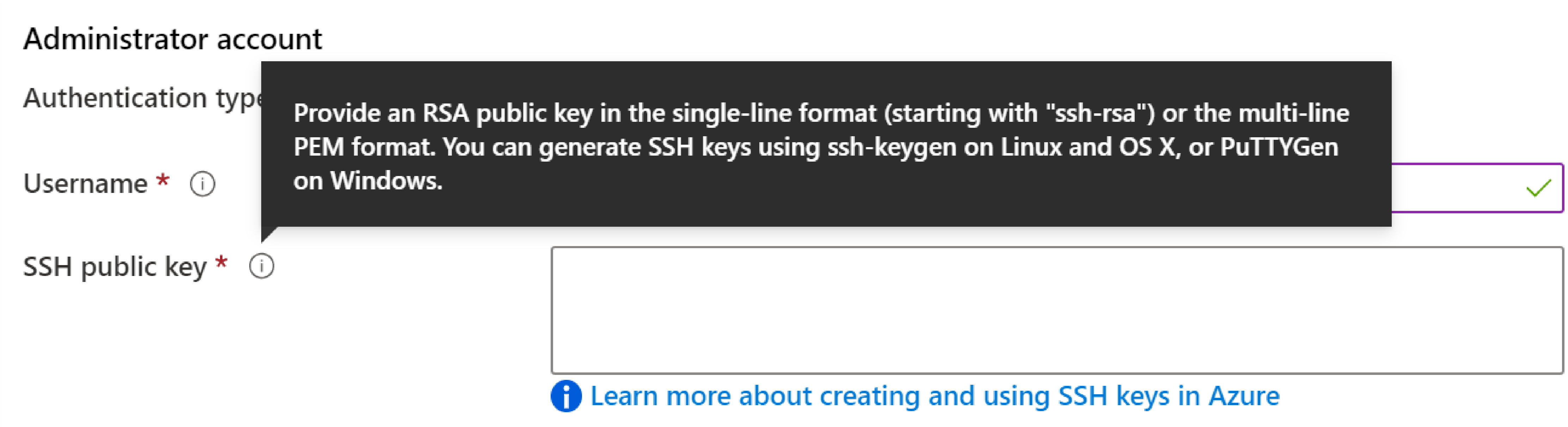
Connect to Windows Virtual Machines

Remote Desktop Protocol (RDP) creates a GUI session and accepts inbound traffic on TCP port 3389

WinRM creates a command-line session so you can run scripts



Connect to Linux Virtual Machines



Authenticate with a SSH public key or password

SSH is an encrypted connection protocol that allows secure logins over unsecured connections

There are public and private keys

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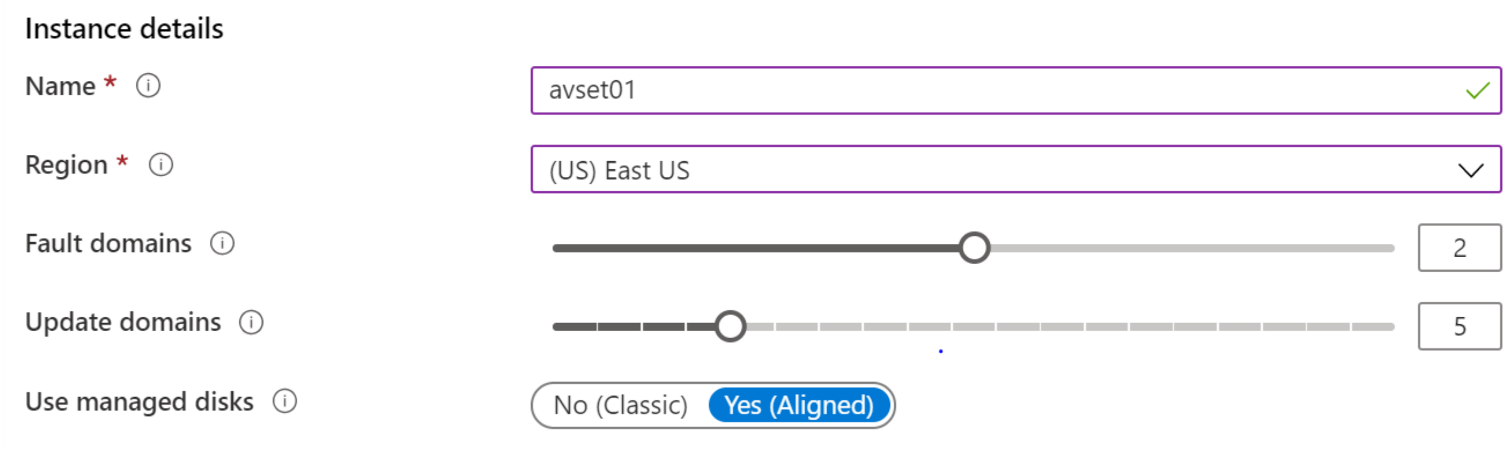
➡️ Deploy and manage Azure compute resources

Configure Virtual Machine Availability

Setup Availability Sets



Two or more instances in Availability Sets = 99.95% SLA

Configure multiple Virtual Machines in an Availability Set	Configure each application tier into separate Availability Sets	Combine a Load Balancer with Availability Sets	Use managed disks with the Virtual Machines

Review Availability Zones

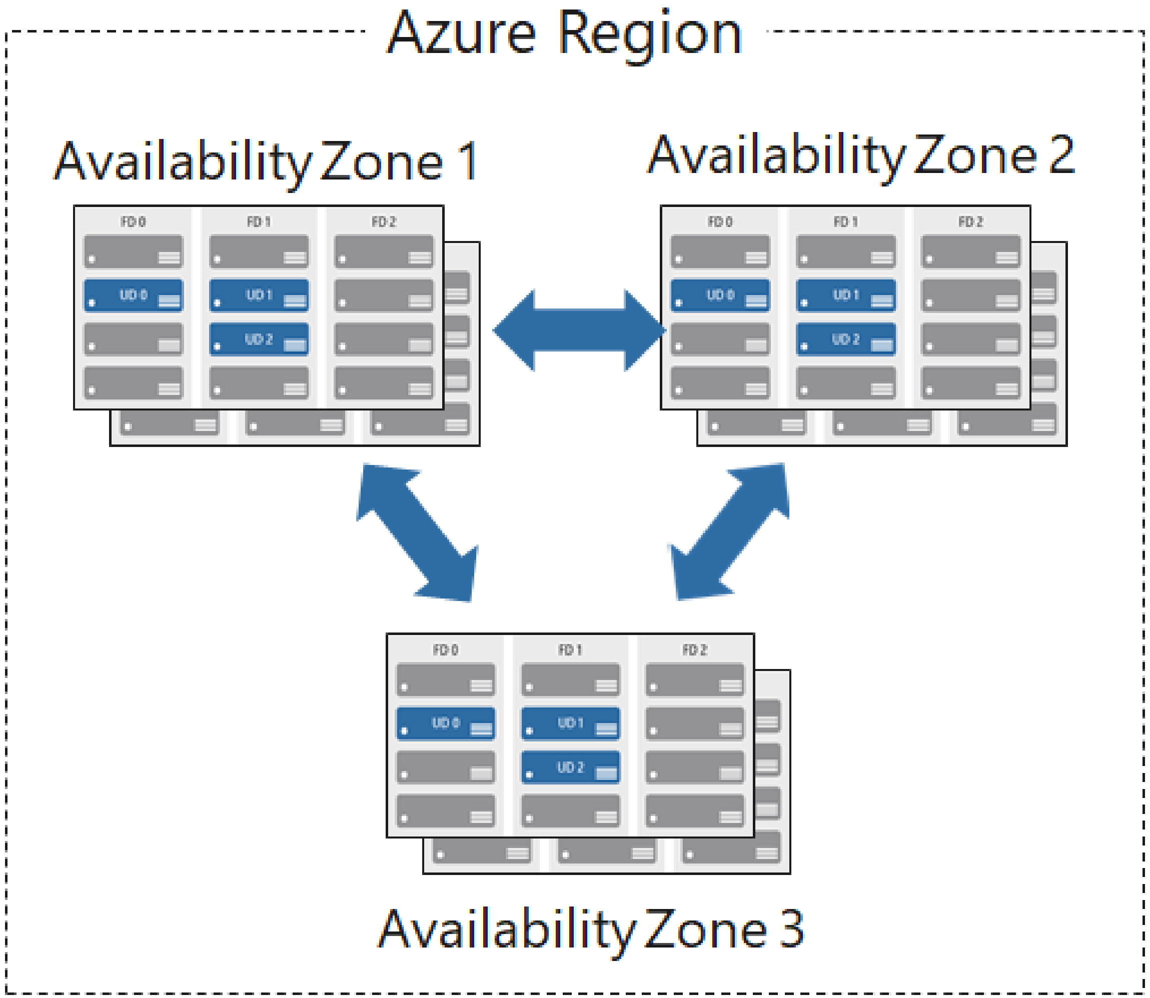
Unique physical locations in a region

Includes datacenters with independent power, cooling, and networking

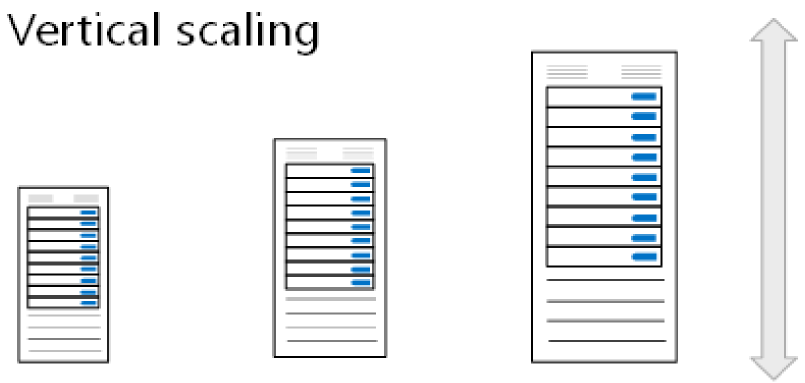
Protects from datacenter failures

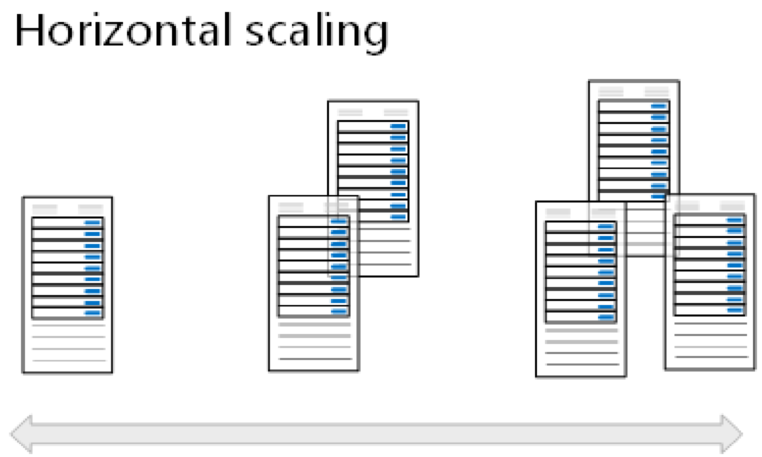
Combines update and fault domains

Provides 99.99% SLA



Compare Vertical to Horizontal Scaling



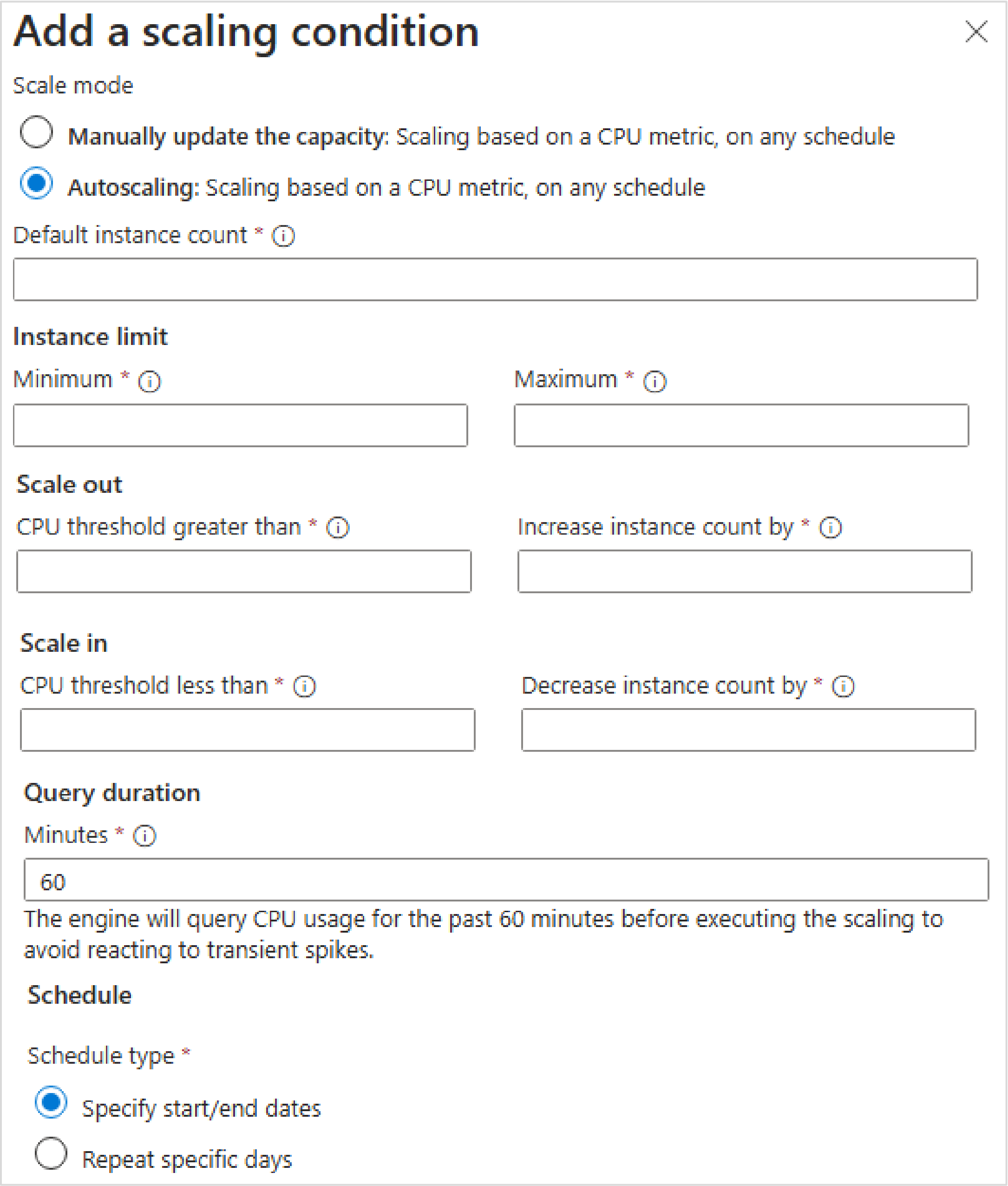


Vertical scaling (scale up and scale down) is the process of increasing or decreasing power to a single instance of a workload; usually manual

Horizontal scaling (scale out and scale in) is the process of increasing or decreasing the number of instances of a workload; frequently automated

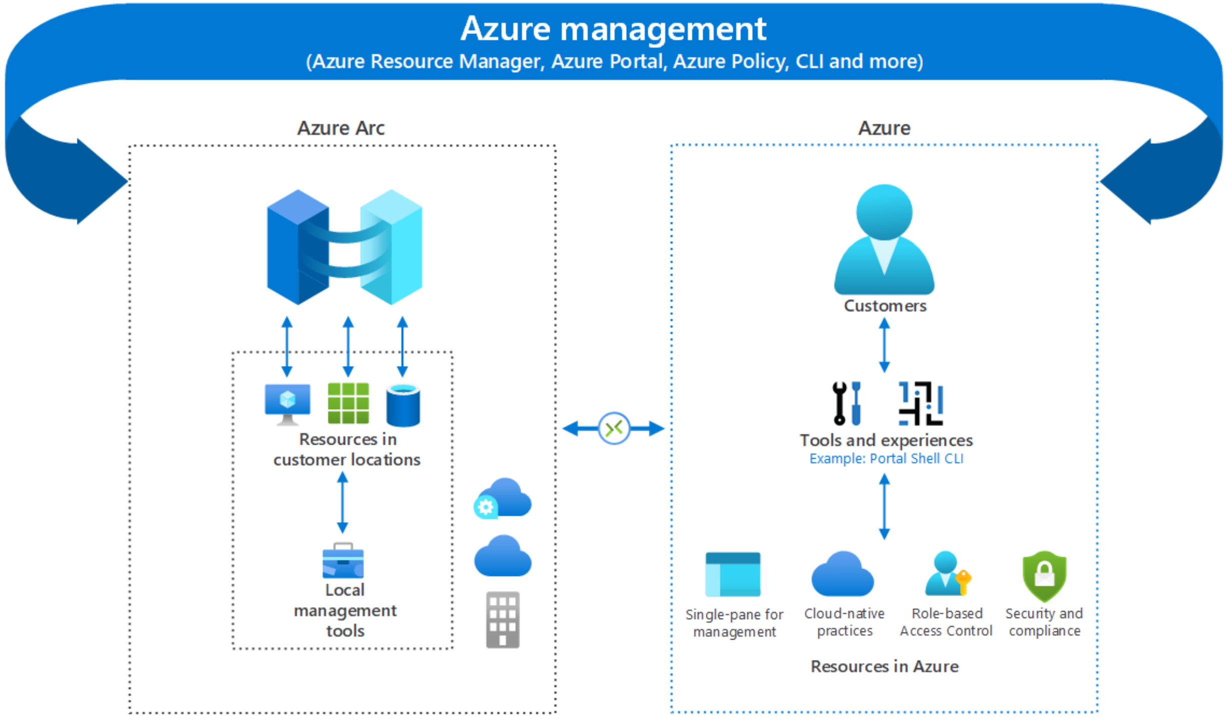
Create and configure scaling

- Manually update or autoscale
- Define a minimum, maximum, and default number of VM instances
- Create more advanced scale sets with scale out and scale in parameters



Bring Azure innovation to your hybrid environments with Azure Arc

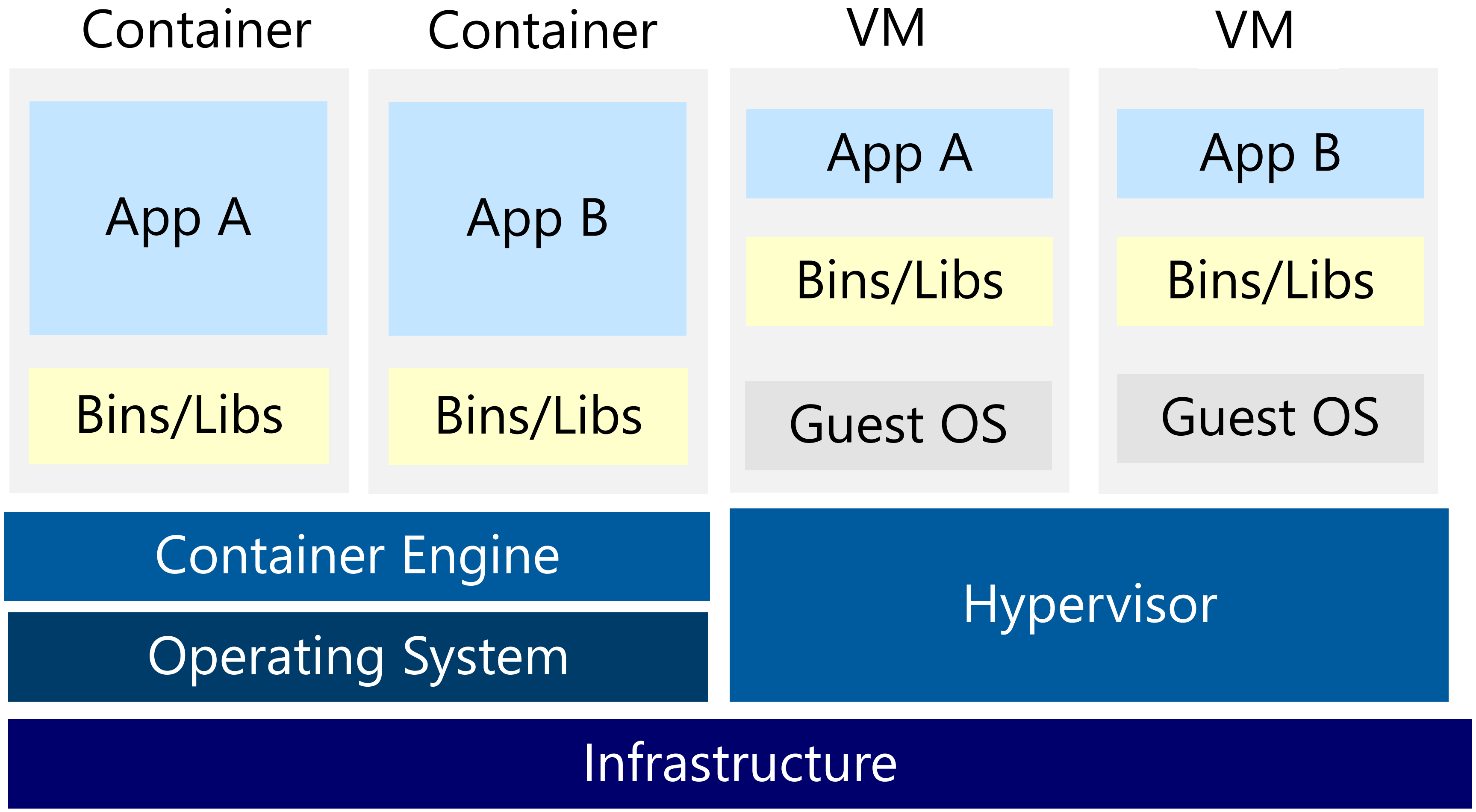
- Manage your entire environment of existing non-Azure and/or on-premises resources
- Manage virtual machines, Kubernetes clusters, and databases as if they are running in Azure.
- Use familiar Azure services and management capabilities, regardless of where your resources live
- Use traditional IT Ops while introducing DevOps practices to support new cloud native patterns in your environment



And now for something completely different…

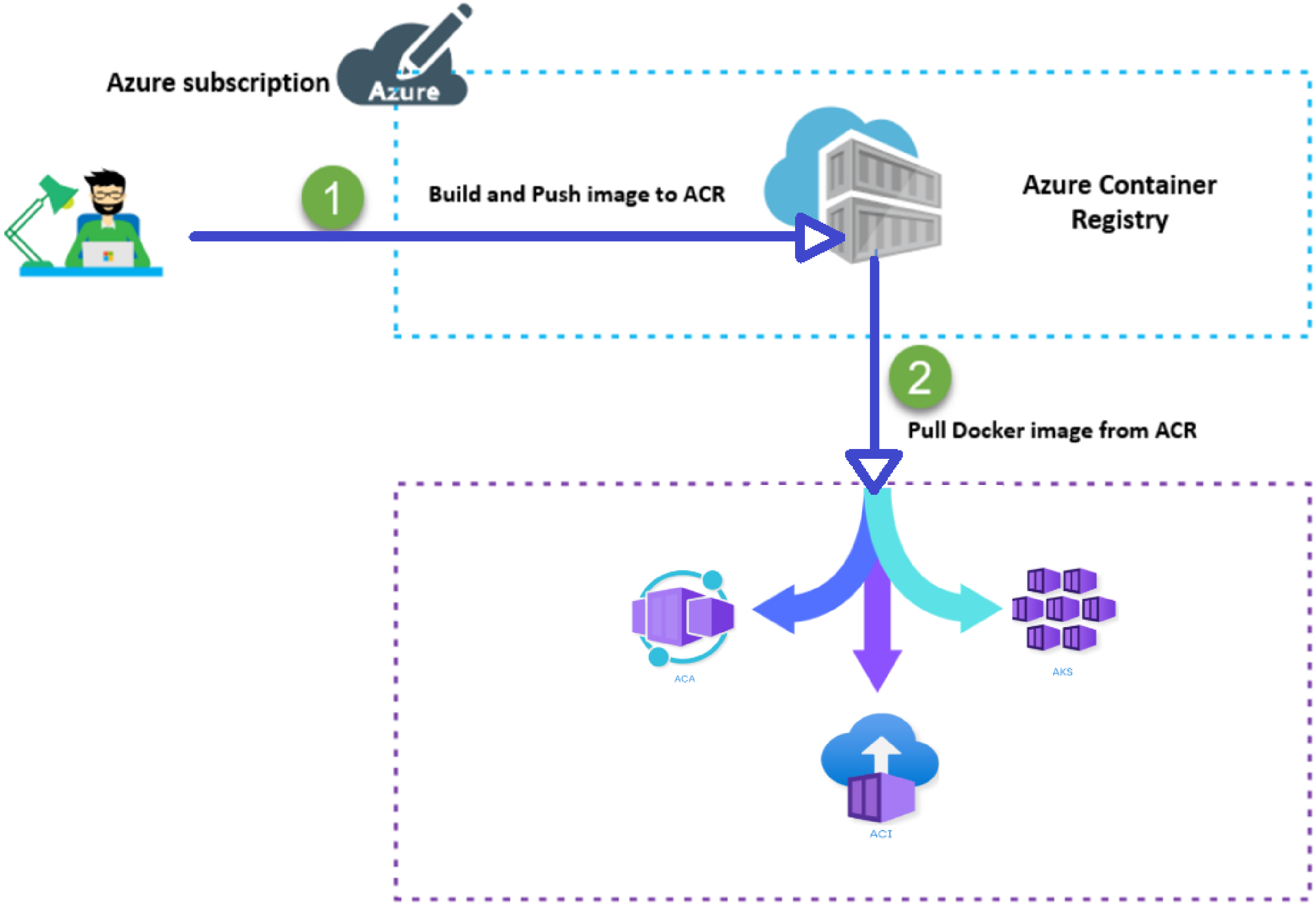
Compare Containers to Virtual Machines

- Isolation
- Operating System
- Deployment
- Persistent storage
- Fault tolerance



Understand Container Images

A container image is a lightweight, standalone, executable package of software that encapsulates everything needed to run an application.

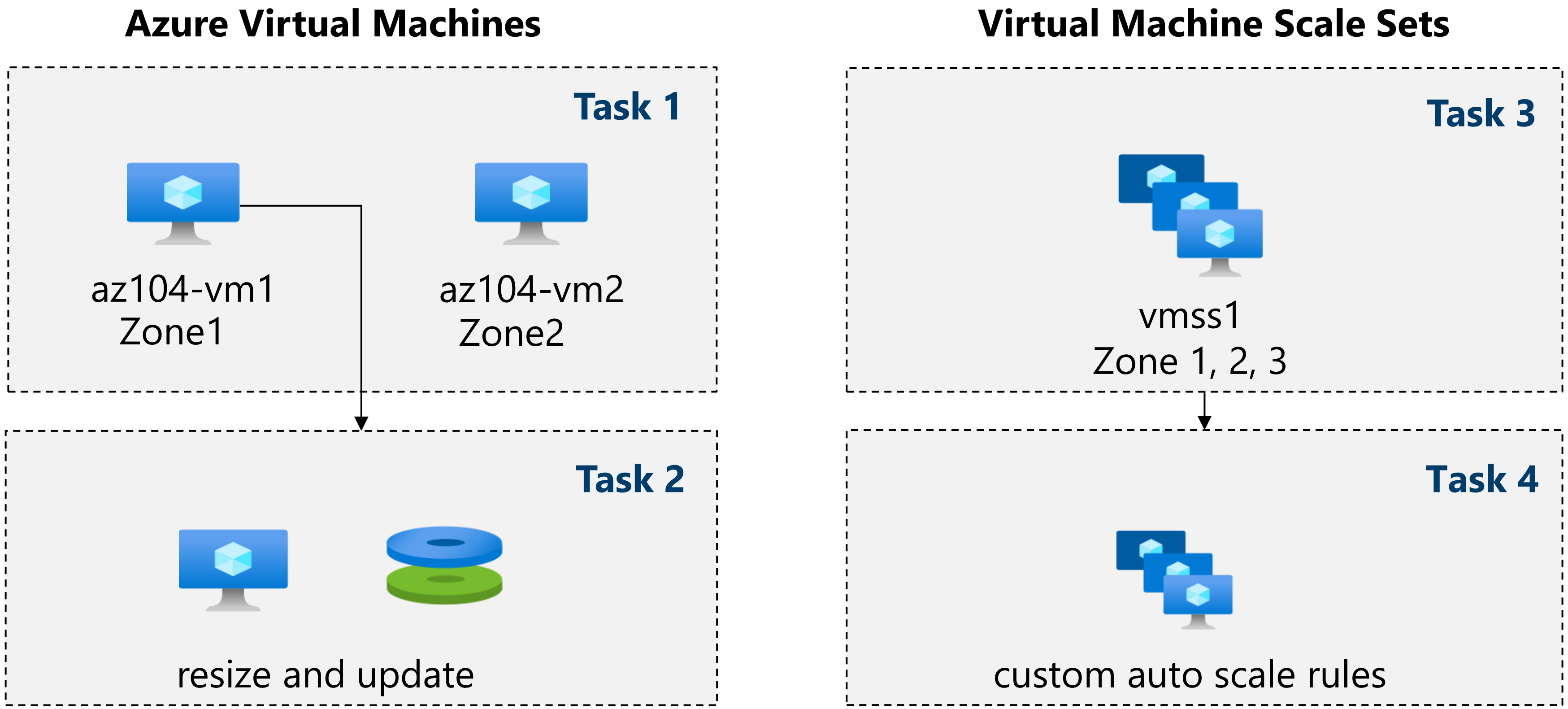


Labs


TO THE LAB ZONE →
NO THROUGH ROUTE
Berlin
Europa
Dublin
SPEEDY ROUTE
4²
tech·ed
lounge
lounge
software diagnostics
E 94
PREWIS
SD Studio
for Team Foundation Server
Visualize your code in
a software map.
software diagnostics
DataCore

Lab 08 – Manage Virtual Machines

Lab 08 – Architecture diagram



Task 5: Create a virtual machine using Azure PowerShell (option 1)
Task 6: Create a virtual machine using the CLI (option 2)

End of presentation